

• Sec-A ($5 \times 2 = 10$).

1. How we can overcome the speed mismatch between CPU and main memory?

Ans The speed mismatch between CPU and main memory is tackled by using cache memory. This acts as a small, super-fast buffer holding frequently accessed data, allowing the CPU to retrieve it much quicker than from main memory.

2. What do you mean by write-through and write-back technique?

Ans Write-through and write-back are techniques for handling data updates in cache memory:

- Write through: when data is updated in the cache, it's simultaneously written to the main memory as well.
- Write back: only the cache is updated initially. The main memory is updated later, often when the cache line is replaced or on a flush command.

3. What are the different types of ROM?

Ans There are four main types of ROM:

(i) MROM (Masked ROM):

(ii) PROM (Programmable ROM):

(iii) EPROM (Erasable Programmable ROM):

(iv) EEPROM (Electrically Erasable Programmable ROM).

4. Differentiate betⁿ. Static RAM and Dynamic RAM.

Ans

SRAM

DRAM

- (i) Static RAM
- (ii) It stores information as long as power is supplied.

- Dynamic RAM
- It stores information as long as the power is supplied or a few milliseconds when the power is switched off.

(iii) SRAM is faster compared to DRAM.	• DRAM provides slow access speeds.
(iv) It does not have a refreshing unit.	• It has a refreshing unit.
(v) These are expensive.	• These are cheaper.
(vi) SRAM are low-density devices.	• DRAM are high-density devices.
(vii) These are used in cache memories.	• These are used in main memories.
(viii) SRAM has higher data transfer rate.	• DRAM has lower data transfer rate.

5. Differentiate between Static EPROM and EPROM.

Ans

Static EPROM	EPROM
• Stand for, static Electrically Erasable Programmable Read Only Memory.	• Erasable Programmable Read-only memory.
• In EPROM, electric signal is used to erase the EPROM's contents.	• In EPROM, UV light is used to erase the EPROM's content.
• Relative size of cell in EPROM is 3.	• Relative size of cell in EPROM is one.
• EPROM is the modern version of EPROM.	• EPROM is the Modern version of PROM.
• The transistor used in EPROM consumes 5 volt.	• The transistor used in EPROM consumes 12.5 volts.
• In EPROM, an erasure consumes 5 milliseconds time for erasing contents.	• In EPROM, an erasure consumes 15 to 20 minutes for erasing contents.
• It is expensive than EPROM.	• It is cheaper than EPROM.

• Sec-B (7x3=21)

1. Explain the Memory techniques in detail.

Ans In computer arch., memory techniques are methods used to optimize the performance and efficiency of memory systems. These techniques aim to reduce memory access times, improve data transfer rates, and enhance overall system performance. Some memory techniques used in computer architecture are:

~~(i) Memory~~ :

(i) Memory Hierarchy :

- principle : organizes memory into levels based on speed and capacity. Faster, smaller memories are closer to the CPU (processor) and slower, larger memories are farther away.

- Levels (typical) :

- Registers :

- Cache

- Main memory (RAM) :

- Secondary Storage

(ii) Cache Memory Techniques :

- Cache Mapping - Determines how data from main memory is mapped to cache (lines / fixed-size blocks).

(a) Direct Mapping : Simplest approach, each memory block maps to a specific cache line.

(b) Associative Mapping - Allows any memory block to be placed in any cache line, offering more flexibility but requires complex hardware for comparisons during lookups.

(ii) Set-Associative Mapping: A compromise, allowing data to be placed in each set but restricting mapping compared to fully associative.

• Cache Replacement Policy: Decides which data to evict from the cache when a new item needs to be placed.

• Least Recently Used (LRU):

• First-In-First-Out (FIFO):

• variations:

• Write Policies: Determine how writes are handled with respect to cache and main memory.

• Write-through: Updates both cache and main memory simultaneously. Offers consistency but can be slower due to double writes.

• Write-back: Updates the cache first, main memory later (on eviction or flush command). Faster for writes but risks data loss if power fails before write-back.

(iii) Main Memory Optimization:

• Memory Interleaving: Divides memory modules into multiple banks and allows concurrent access to different banks, improving overall memory bandwidth.

• Error Correction Code (ECC): Technique for detecting and correcting memory errors during data access.

(iv) Memory Addressing:

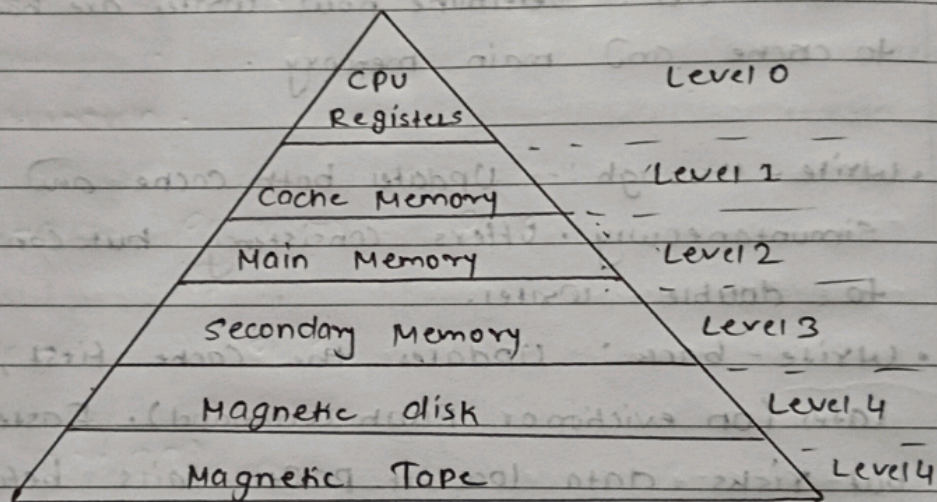
• Technique: Memory addresses specify the location of data in memory. Techniques like virtual memory and memory

mapped I/O extend the capabilities of physical memory addressing.

- Virtual Memory :- creates the illusion of having more memory than physically available by utilizing secondary storage.

Q2. Explain the memory Hierarchy in detail.

Ans. The memory hierarchy refers to the organization of memory subsystem in a computer system, with each level offering different characteristics in terms of speed, capacity, cost and volatility.



(i) Registers: Registers are small high-speed memory units located in the CPU. They are used to store the most frequently used data and instructions. Registers have the fastest access time and the smallest storage capacity typically ranging from 16 to 64 bits.

(ii) Cache Memory :- Cache memory is a small, fast memory unit located close to the CPU. It stores frequently used data and instructions that have been recently accessed from the main memory.

(iii) Main Memory :- Main memory also known as RAM, is the primary memory of a computer system. It has a larger storage capacity than cache memory, but it is slower.

(iv) There are two types of Main Memory:

(i) Static RAM

(ii) Dynamic RAM

(iv) Secondary Storage - Secondary storage, such as HDD, SSD is a non-volatile memory unit that has a larger storage capacity than main memory. It is used to store data and instructions that are not currently in use by CPU.

(v) Magnetic Disk :- Magnetic disk are simply circular plates that are fabricated with either a metal or a plastic or a magnetized material.

(vi) Magnetic Tape: Magnetic Tape is simply a magnetic recording device that is covered with a plastic film. It is generally used for the backup of data.

Q3. Define Hit Ratio with suitable example.

Ans. A hit ratio also known as Cache Hit Ratio, is a measurement of how many content requests a cache is able to fill successfully, compared to how many requests it receives.

Ans. The hit ratio, also known as cache hit ratio, refers to the ratio of cache accesses that result in a cache hit to the total number of cache accesses.

Example:

→

Formula to calculate the hit ratio is:

$$\text{Hit Ratio} = \frac{\text{Number of cache hits}}{\text{Total No. of Memory Accesses}} \times 100\%$$

ex:

Suppose we have a system with cache memory and a main memory. over a period of time, the system performs a total of 100 memory accesses. out of these accesses, 80 requests find the data in the cache (cache hits), while the remaining 20 requests require fetching the data from main memory (cache miss).

Using the formula:

$$\text{Hit Ratio} = \frac{80}{100} \times 100 = 80\%$$

The hit ratio is 80%.

• Sec-c (11x3=33)

1. Explain the address mapping of the RAM/ROM in detail by using the appropriate example.

Ans Address mapping in RAM and ROM involves the process of translating logical memory addresses generated by the CPU into physical addresses that correspond to specific locations in the memory module. The mapping is crucial for the proper functioning of the memory subsystem and ensures that data can be accessed and stored correctly.

• Address Mapping in RAM:

⑥ Physical RAM Module:

RAM is Volatile memory used for temporary storage of data that a computer's CPU needs to access quickly. The address mapping of RAM is typically straightforward and follows a linear addressing scheme.

For ex: Let's consider a hypothetical computer system with 8 GB of RAM. In this system, each byte of RAM is assigned a unique address. The addresses range from 0 to 8,589,934,591 (assuming 1 byte addressing). Each address corresponds to a specific memory location in the RAM.

- Address 0: Byte 0

- Address 1: Byte 1

- Address 2: Byte 2

- Address n: Byte n

This linear mapping allows the CPU to access any byte of RAM by specifying its unique address. For instance, if the CPU needs to read data from address 1000, it can directly access the byte stored at that address without needing to traverse through other memory locations.

- Address Mapping of ROM: ROM is non-volatile memory used to store firmware or software that is not expected to be modified during normal operation. The address mapping of ROM can vary depending on the type

of ROM, such as Mask ROM, PROM, EPROM, or EEPROM.

Let's an example, of Computer system with a ROM chip containing the system's BIOS (Basic Input/output system). BIOS is essential software stored on ROM that initializes hardware components and loads the operating system during the boot-up process.

In this example, the ROM chip might be 4MB in size, and it follows a linear address mapping similar to RAM. Each byte of ROM is assigned a unique address, ranging from 0 to 33,554,431 (assuming 1 byte addressing).

- Address 0 : Byte 0 (contains BIOS code)
- Address 1 : Byte 1
- Address 2 : Byte 2
- - - - -
- - - - -
- Address n : Byte n

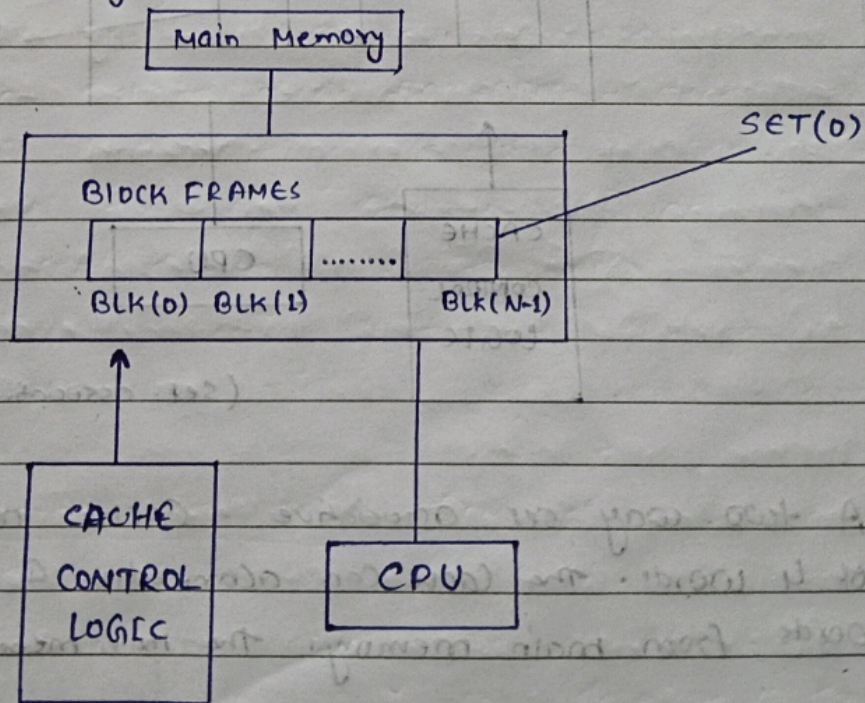
Unlike, RAM, ROM contents are typically fixed and cannot be modified by the CPU during normal operation. Therefore, the address mapping of ROM remains static, and the CPU can access the stored firmware or software instructions directly using their respective addresses.

Q2. Explain the concept of Associative and Set-associative mapping in Cache memory.

Ans Cache memory is a special very high-speed memory. The cache is a smaller and faster memory that stores copies of the data from frequently used main memory locations.

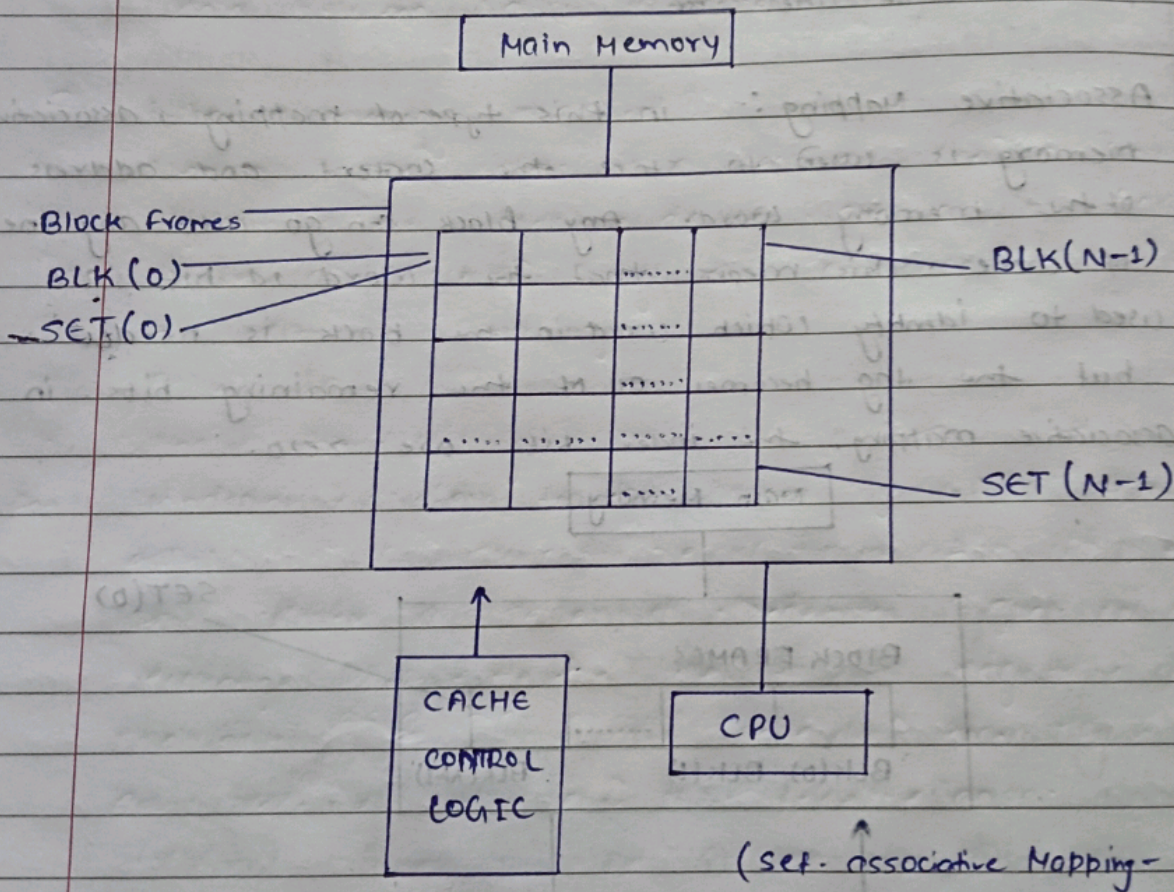
• Cache Mapping :- There are three types of mapping used for the purpose of cache memory.

- (i) Direct Mapping - The simplest technique, known as direct mapping, maps each block of main memory into only one possible cache line.
- (ii) Associative Mapping :- In this type of mapping, associative memory is used to store the content and address of the memory word. Any block can go into any line of cache. This means that the word id bits are used to identify which word in the block is needed, but the tag becomes all of the remaining bits. In associative mapping, the index bits are zero.



(Associative Mapping - Structure)

(iii) Set-Associative Mapping :- This form of mapping is an enhanced form of direct mapping where the drawbacks of direct mapping are removed. Set associative addresses the problem of possible thrashing in the direct mapping method. It divides the cache into a number of sets, each containing multiple cache lines. When a block of data needs to be cached, it can be placed in any cache line within a specific set determined by a set index. Within each set, associative mapping is used to determine which cache line the block will occupy.



Q3. A two-way set associative cache memory uses block of 4 words. The cache can accommodate a total of 2048 words from main memory. The main mem size is $128K \times 32$.

(i) Formulate all pertinent information required to construct the cache memory.

(ii) What is the size of cache memory and number of address bits required to define a block of Main Memory?

→ Cache Configuration -

Set associative cache with two ways each block have 4 words.

→ Cache Capacity -

Total : 2048 words

→ Main Memory Configuration

Main Memory size : $128K \times 32$

Each word in main memory is 32 bits size.

(i) Size of Cache memory -

$$\text{Total no. of cache block} = \frac{2048}{4} = 512 \text{ blocks}$$

$$\text{Total no. of sets} = \frac{512 \text{ block}}{2 \text{ way}} = 256 \text{ sets}$$

$$\text{Size of Cache memory} = 256 \text{ sets} \times 2 \text{ ways} \times 4 \text{ words} \times 32 \text{ bit}$$

$$= 32768 \text{ bits}$$

$$= 4KB$$

(ii) No. of Address Bits Required -

$$\text{No. of Address bits} = \log_2 (128K \times 32)$$

$$= \log_2 (128 \times 1024 \times 32)$$

$$= 22 \text{ bits}$$